

Mathematics

Paper : I

SWL-11-23

Roll No.

--	--	--	--	--	--

(To be filled in by the candidate)

H.S.S.C (11th) 1st Annual 2023

Time : 30 Minutes

Objective – (iii)

Marks : 20

Paper Code

6

1

9

5

Note : – You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

SECTION-A

Q.1	Questions	A	B	C	D
1.	$\cos^2 \alpha - \sin^2 \alpha =$	1	$\cos 2 \alpha$	$\sin 2 \alpha$	$\sin 3 \alpha$
2.	$\cos(\sin^{-1} \frac{1}{\sqrt{2}}) =$	$\frac{1}{\sqrt{2}}$	$\frac{-1}{\sqrt{2}}$	$\frac{1}{2}$	$\frac{-1}{2}$
3.	$r_1 =$	$\frac{s}{s-a}$	$\frac{\Delta}{s-a}$	$\frac{\Delta s}{s-a}$	$\frac{\Delta}{s+a}$
4.	The expansion $1+x+x^2+x^3+\dots+x^r+\dots =$	$(1+x)^{-1}$	$(1-x)^{-1}$	$(1+x)^{-2}$	$(1+x)^{-3}$
5.	The period of cosec θ is.	2π	π	$\frac{\pi}{2}$	$\frac{\pi}{6}$
6.	Solution of $\cot \theta = \frac{1}{\sqrt{3}}$ which lies in $[0, 2\pi]$	$\frac{\pi}{3}$	$\frac{\pi}{4}$	$\frac{\pi}{6}$	$\frac{\pi}{2}$
7.	Value of $(-1)^{\frac{-21}{2}}$ is.	-1	1	i	$-i$
8.	The statement $n! > 2^n - 1$ is true for	$n=1$	$n=2$	$n < 4$	$n \geq 4$
9.	$\frac{a}{2 \sin \alpha} =$	r	r_1	Δ	R
10.	$n(n-1)(n-2)\dots(n-r+1) =$	$\frac{n!}{r!}$	$\frac{n!}{(n-r)!}$	$\frac{n!}{(n+r)!}$	nC_r
11.	A square matrix $A=[a_{ij}]$ is called hermitian matrix if:	$A^t=A$	$A^t=-A$	$(\bar{A})^t=A$	$(\bar{A})^t=-A$
12.	Next two terms of sequence 7, 9, 12, 16, ... are:	21, 27	21, 26	20, 27	20, 26
13.	Formula for sum of an infinite geometric series is.	$a_1 + (n-1)d$	$\frac{a_1}{1-r}$	$\frac{a_1(1-r^n)}{1-r}$	$\frac{a_1(r^n-1)}{r-1}$
14.	The rational fraction $\frac{x^2+1}{x^3-1}$ is:	Identity	Irrational	Proper	Improper
15.	$\omega^{28} + \omega^{29} + 1 =$	-1	0	1	2
16.	Which is an exponential equation?	$x^2+1=0$	$x^3+1=0$	$2x+1=0$	$2^x-1=0$
17.	Value of $\frac{10!}{7!}$ is:	718	719	720	730
18.	If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $ A =$	$ad+bc$	$bc-ad$	$ad-bc$	$ac-bd$
19.	The number of elements in power set of $\{0,1\}$ are.	4	3	2	1
20.	$1+\cot^2 \theta =$	$\cos^2 \theta$	$\sin^2 \theta$	$\operatorname{cosec}^2 \theta$	$\sec^2 \theta$

CANCELLED

11-323-14-18500 ★★★

Mathematics

Paper : I

SWL-11-23

Roll No.

--	--	--	--	--	--	--

(To be filled in by the candidate)

H.S.S.C (11th) 1st Annual 2023

Time : 2:30 Hours

Subjective

Marks : 80

Note: - Section B is compulsory. Attempt any Three questions from section C.

SECTION - B

2. Write short answers to any Eight parts.

(8 x 2 = 16)

- Factorize $3x^2 + 3y^2$.
- Prove that $\bar{z} = z$ iff z is real.
- What is the difference between $\{a, b\}$ and $\{\{a, b\}\}$?
- Show that the statement is a tautology: $\sim q \wedge (p \rightarrow q) \rightarrow \sim p$
- Define Monoid.
- If $A = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$, then show that $4A - 3A = A$.
- Find the inverse of matrix $\begin{bmatrix} 2i & i \\ i & -i \end{bmatrix}$.
- Write any two properties of determinants.
- Solve the equation by using quadratic formula: $15x^2 + 2ax - a^2 = 0$.
- Define a reciprocal equation and give one example.
- Prove that $\left(\frac{1+\sqrt{-3}}{2}\right)^9 + \left(\frac{1-\sqrt{-3}}{2}\right)^9 = -2$
- Discuss the nature of roots of equation $x^2 + 2x + 3 = 0$.

3. Write short answers to any Eight parts.

(8 x 2 = 16)

- Define proper rational fraction.
- Find a_2 and a_3 of the sequence in which $a_n = na_{n-1}$ and $a_1 = 1$
- Which term of A.P; 5, 2, -1, ... is -85.
- Sum the series $\frac{1}{1+\sqrt{x}} + \frac{1}{1-x} + \frac{1}{1-\sqrt{x}} + \dots$ to n terms.
- Define harmonic mean between a and b . Write its formula also.
- Find the sum to n term of series whose n th term is $2n+3$.
- Write $n(n-1)(n-2)\dots(n-r+1)$ into factorial form.
- Prove that ${}^nC_r = {}^nC_{n-r}$.
- Calculate number of diagonals of 5 sided figure.
- Evaluate $(9.9)^5$
- Find middle term in the expansion of $\left(\frac{1}{x} - \frac{x^2}{2}\right)^{12}$
- Expand $(8-5x)^{-2/3}$ upto two terms.

SWL-11-23

(2)

4. Write short answers to any Nine parts.

(9 x 2 = 18)

- i. Define the word 'Trigonometry'.
- ii. Convert 3 radians into degree.
- iii. Find $\sin \theta$ and $\cos \theta$ when $\theta = \frac{-7\pi}{4}$.
- iv. Express $\cos 7\theta - \cos \theta$ as product form.
- v. Find the value of $\sin 2\alpha$ when $\sin \alpha = \frac{12}{13}$, where $0 < \alpha < \frac{\pi}{2}$.
- vi. Find the value of $\tan 105^\circ$.
- vii. Find the period of $\cos \frac{x}{6}$.
- viii. Solve the right triangle, in which $\alpha = 58^\circ 13'$, $b = 125.7$ and $\gamma = 90^\circ$.
- ix. Write half angle formulas for $\sin \frac{\alpha}{2}$ and $\sin \frac{\beta}{2}$.
- x. By using the cosine and sine law, solve the triangle ABC given that $b=3, c=5, \alpha = 120^\circ$.
- xi. Find the value of $\tan^{-1}(-\sqrt{3})$.
- xii. Define trigonometric equation.
- xiii. Find the solution of $\sec x = -2$ which lies in the interval $[0, 2\pi]$.

SECTION - C

Note: Attempt any Three questions. Each question carries (5+5=10) marks.

(10 x 3 = 30)

5. (a) Find the value of x if $\begin{vmatrix} 3 & 1 & x \\ -1 & 3 & 4 \\ x & 1 & 0 \end{vmatrix} = -30$

(b) Show that the roots of $x^2 + (mx + c)^2 = a^2$ will be equal, if $c^2 = a^2(1 + m^2)$.

6. (a) Resolve $\frac{x^2 + 1}{x^3 + 1}$ into partial fraction.

(b) Two dice are thrown twice. What is the probability that sum of dots shown in the first throw is 7 and that of second is 11?

7. (a) Find 'n' so that $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ may be the A.M. between a and b .

(b) Use mathematical induction to prove that $x+y$ is a factor of $x^{2n-1} + y^{2n-1}$, $x \neq -y$.

8. (a) Prove that $\sin^6 \theta - \cos^6 \theta = (\sin^2 \theta - \cos^2 \theta)(1 - \sin^2 \theta \cdot \cos^2 \theta)$.

(b) If $\alpha + \beta + \gamma = 180^\circ$, show that $\cot \alpha \cdot \cot \beta + \cot \beta \cdot \cot \gamma + \cot \gamma \cdot \cot \alpha = 1$

9. (a) Solve the triangle ABC in which $a = \sqrt{3} - 1$, $b = \sqrt{3} + 1$ and $\gamma = 60^\circ$.

(b) Prove that $\sin^{-1} \frac{77}{85} - \sin^{-1} \frac{3}{5} = \cos^{-1} \frac{15}{17}$.

CANCELLED

215-323-14-18500